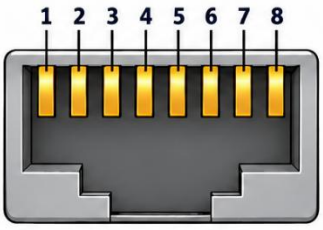
Assignment – 1.5

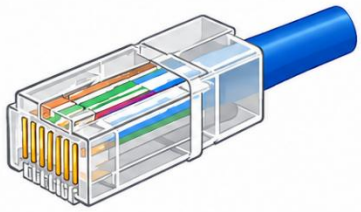
Ethernet standard and specification

Q - 1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

RJ-45 Connector Pin Layout



Front view (looking into the connector)



RJ-45 plug (with wires)

Color Code Legend

- White/Orange (WO)
- Orange (O)
- White/Green (WG)
- Green (G)
- White/Blue (WB)
- Blue (B)
- White/Brown (WBR)
- Brown (BR)

T568A Wiring Standard

Pin No.	Color Code	Wire
1	White/Green (WG)	White with Green stripe
2	Green (G)	Solid Green
3	White/Orange (WO)	White with Orange stripe
4	Blue (B)	Solid Blue
5	White/Blue (WB)	White with Blue stripe
6	Orange (O)	Solid Orange
7	White/Brown (WBR)	White with Brown stripe
8	Brown (BR)	Solid Brown

T568B Wiring Standard

Pin No.	Color Code	Wire
1	White/Orange (WO)	White with Orange stripe
2	Orange (O)	Solid Orange
3	White/Green (WG)	White with Green stripe
4	Blue (B)	Solid Blue
5	White/Blue (WB)	White with Blue stripe
6	Green (G)	Solid Green
7	White/Brown (WBR)	White with Brown stripe
8	Brown (BR)	Solid Brown

<u>Q - 2.</u> Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable Type	Cat 3 or higher twisted-pair cable	Cat 5 or higher twisted-pair cable	Cat 5e or higher twisted-pair cable
Maximum Cable Length	100 meters	100 meters	100 meters
Connector Type	RJ-45	RJ-45	RJ-45
Number of Pairs Used	2 pairs	2 pairs	4 pairs
Common Usage	Older Ethernet networks	Fast Ethernet networks	Gigabit Ethernet networks

<u>Q - 3.</u> Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why. Hint: Think about how transmit and receive pairs are wired in each cable type.

➢ Answer: Crossover Cable

To connect two laptops directly using Ethernet, a **crossover cable** is traditionally used.

➢ Reason:

- A crossover cable swaps the transmit (TX) and receive (RX) wire pairs.
- The transmit pins of one laptop connect to the receive pins of the other laptop, and vice versa.
- This allows both laptops to send and receive data directly without a network switch or hub.

Q - 4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

No.	Device	Likely Ethernet Standard	Speed
1	Router	1000BASE-T	1 Gbps
2	Desktop Computer	1000BASE-T	1 Gbps
3	Network Printer	100BASE-TX	100 Mbps